

Metallography of T3 Aluminum, 1018 Steel, 360 Brass, and Cast Iron

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To: Dr. Terry Xu

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Equipment and Procedure

In the Metallography experiment the first step was to sand the surface of the sample material. The sanding had three to four stages depending on the sample hardness. If the sample was a hard material such as 1018 steel, then the first step would be to sand with a 240-grit piece of sandpaper until the previous surface lines were replaced with a new surface. This would be done four times going from 240 grit to 320 grit to 400 grit and finally to 600 grits. For softer samples such as Aluminum only three steps were required because the 240-grit stage was redundant. After the surfaces were verified, the next step was to use the Ecomet Polishers to polish the material's surface. Depending on the sample hardness, different grit sizes were utilized. For softer samples such as the T3 Aluminum, a grit size of 1200 was needed and was run at 100 RPM. Inversely, harder samples such as the 1018 Steel required a grit size of 800 and was run at 200 RPM. The first part of polishing used water and the aforementioned grit sizes while the second part required chemical polishing to be performed. The second part of polishing used Al_2O_3 and a microfiber polishing pad. The samples were then brought to the Olympus BX51 Microscope to be inspected for any large scratches that might have remained before moving on to etching. When the sample was successfully inspected, the next step was to etch the material to reveal the micro surface. Etching was done by dipping the surface of the sample into an etching agent for a set amount of time. After the etching was done, the sample was then moved to the Olympus BX51 Microscope to capture the image of the sample's microstructure.

Result and Discussion

Microscopic pictures are taken in order to analyze the microstructure of three different samples, 1018 Steel, T3-Aluminum, and Cast Iron. Figure 1 is a picture of the microstructure of T3-Aluminum at 100x magnification.

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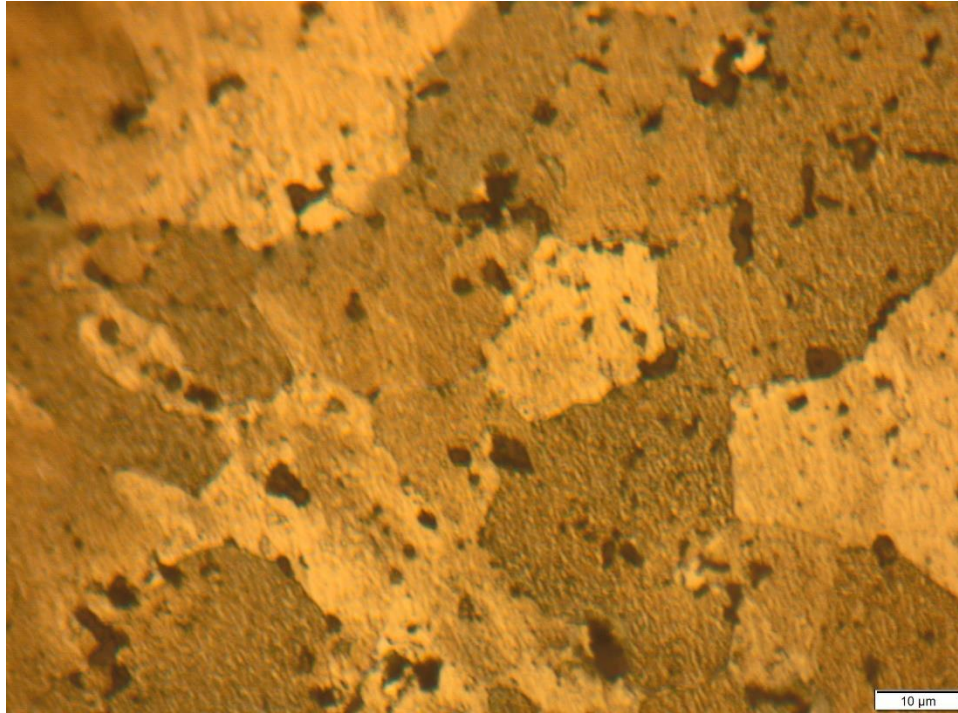


Figure 1. Microstructure of T3-Aluminum at 100x Magnification Taken by Mikey Wiesecke using an Olympus BX51 Microscope

As seen in the picture, the T3-Aluminum has two types of impurities; insoluble impurities and soluble impurities. The soluble impurities, which are often iron or magnesium, are the black areas in the pictures while the insoluble impurities, copper, and copper Magnesium, are the open circles in the picture¹. The Impurities distinguish the mechanical properties from the expected properties, but these are unfortunately unavoidable especially those which are insoluble impurities. Furthermore, the soluble impurities are a sign that the aluminum can be heat treated to try to negate some of the effects from these impurities.

The next figure, Figure 2, is a picture of 360 Brass at 50x magnification.

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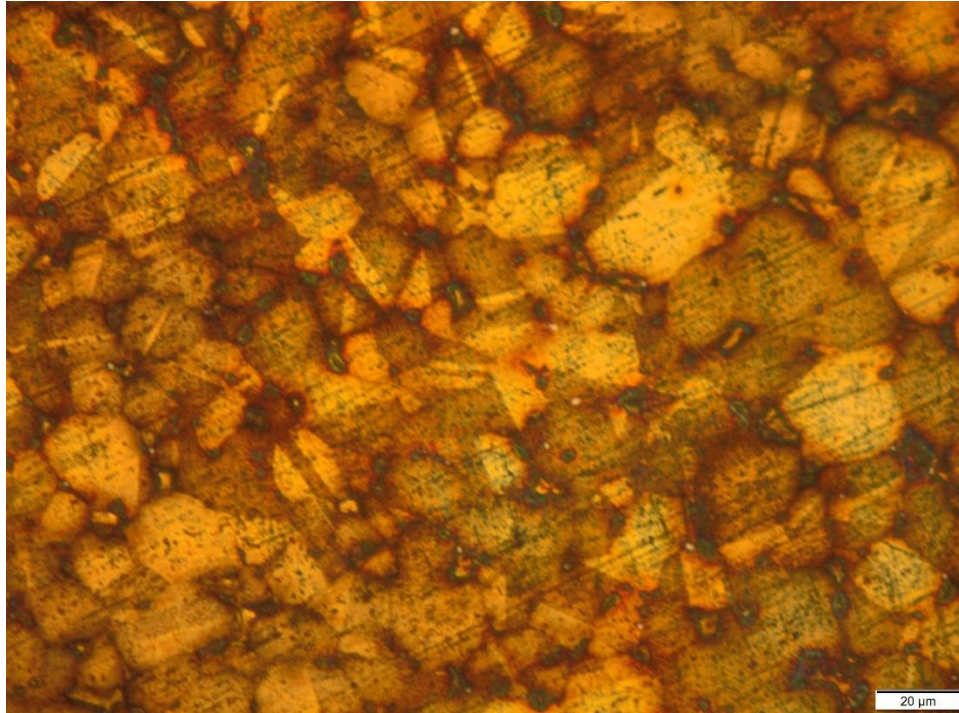


Figure 2. Microstructure of 360 Brass at 50x magnification taken by Ishen Thakur

Brass is composed of two materials, copper, and zinc. It displays a two-phase structure with an alpha phase and a beta phase. The yellow regions are the alpha phase, or the copper part while the dark regions are the beta phase, or the zinc part of brass. The black parts of the pictures are different precipitants or particles that are in the material. The next two materials are 1018 Steel and Cast Iron which are shown in Figure 3 and Figure 4 respectively.

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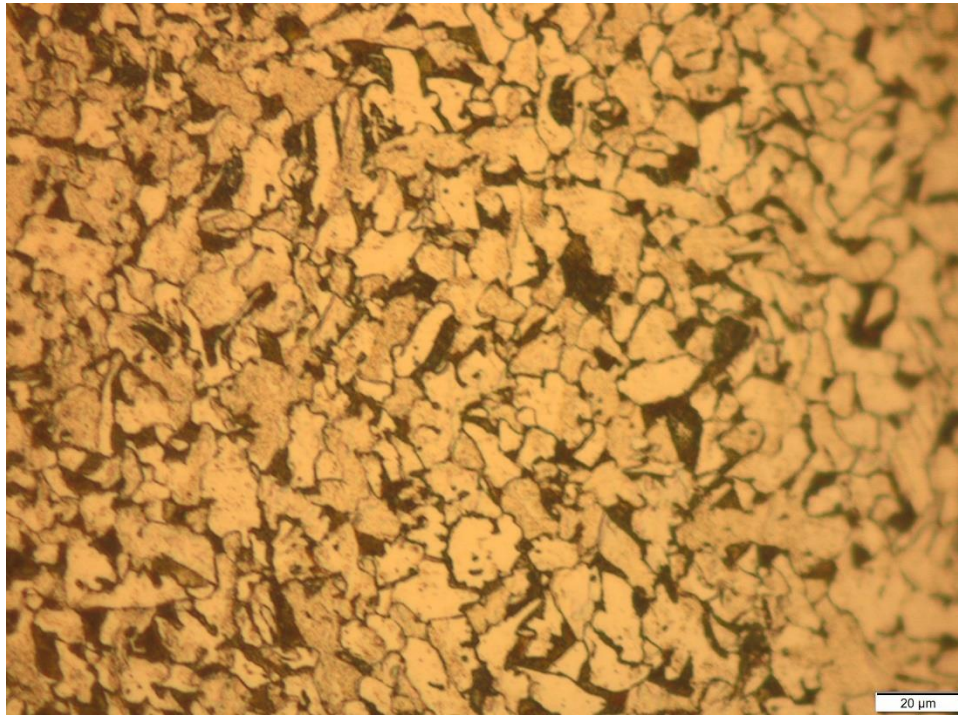


Figure 3. Microstructure of 1018 Steel at 50x Magnification Taken by JWhit

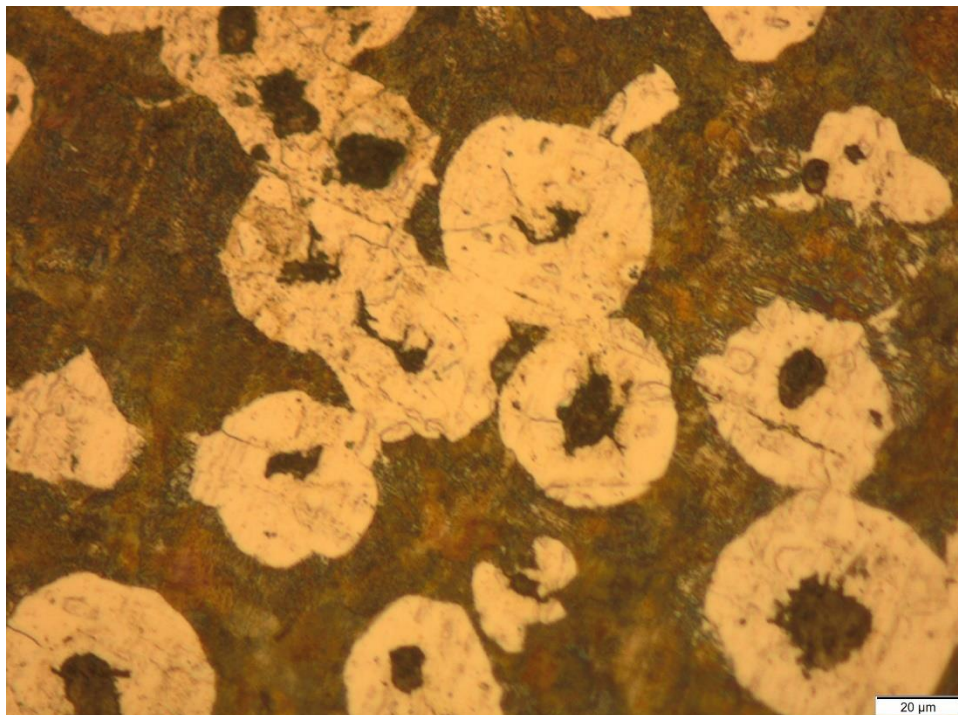


Figure 4. Microstructure of Cast Iron at 50x Magnification taken by TSubba

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1018 Steel and Cast Iron are fundamentally the same, both being made from the combination of iron and carbon. The major difference comes from the concentration of these elements as well as the form of the iron in these materials. In Figure 3, the grain boundaries are very clear, while the grain boundaries in Figure 4 are near impossible to spot. This lends to the fact that 1018 Steel has bigger grains and is more ductile than cast iron whose grains are very difficult to spot at the current magnification. Another key difference between the two steel structures is the structure of the Iron element. When working with iron, there are 4 forms that can result in order from softest to hardest, Ferrite, Pearlite, Cementite and Martensite. The 1018 steel structure is Ferrite dominant and is identified by the lighter yellow area while the darker/black area are Pearlite. Compare this to the Cast Iron which is more Pearlite dominant and it can be concluded that the cast iron is will be stronger because of the presence of more pearlite. Another interesting point from the cast iron is that the Black flakes are surrounded by a spherical lighter area. These are graphite flakes surrounded by ferrite and are key identifier of this Cast Iron being Spherical Graphite Cast Iron. In the Cast Iron, the shape of the graphite can be described as nodular whose shape makes the material much more ductile and tougher than other cast irons.²

Grain Size Based on Microstructure

When looking at the Microstructure of a material the grain structure can be determined with the intercept method. This method involves counting the intercepts/borders between each grain for a set scale. Figure 5 shows the microstructure of 1018 steel with lines applied to count the intercept by the scale.

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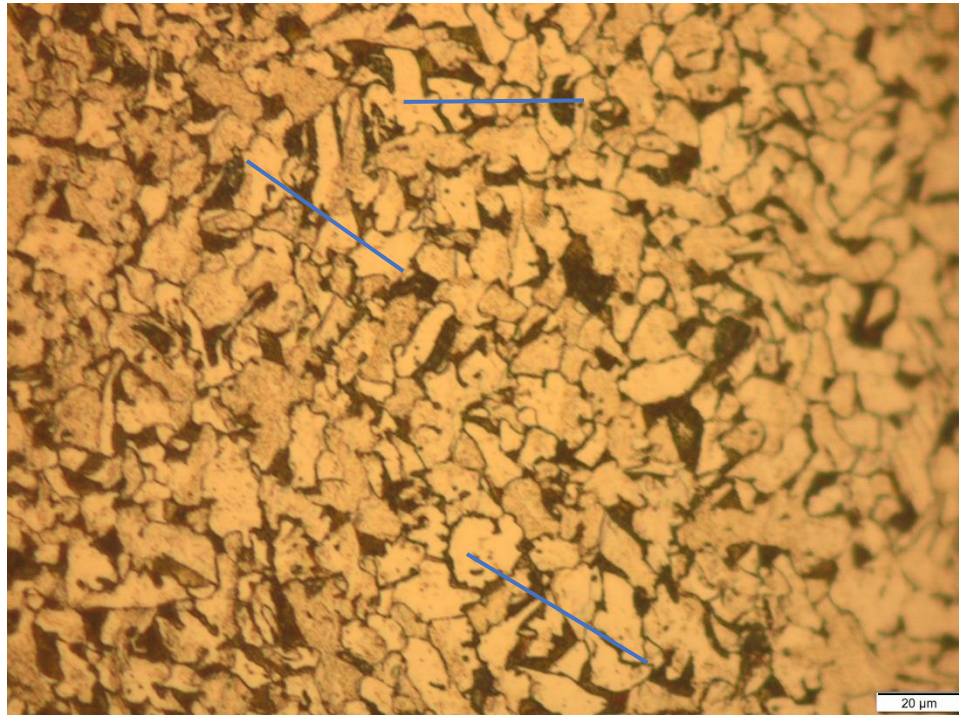


Figure 5

The average number of borders amongst the three lines, using a scale of 40 micrometers, is an intercept count of 6.7. The scale is then divided by the intercept count to get a mean diameter of 5.97 micrometers. Last an Eq.1 ASTM Grain Number Equation is used to calculate the final grain number of 11.48.

$$\text{Eq.1}^3 \quad G = -6.6457 \log(d * 10^3) - 3.298$$

References

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